

MCQ: Metabolic Biochemistry

Cardiac marker, plasma enzyme, plasma protein, screening for metabolic diseases.

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1. Which organ synthesizes most plasma proteins?
A. Kidney
B. Liver
C. Spleen
D. Pancreas
2. What is the main plasma protein responsible for maintaining osmotic pressure?
A. Globulin
B. Albumin
C. Fibrinogen
D. Hemoglobin
3. Which plasma protein functions in blood coagulation?
A. Albumin
B. Fibrinogen
C. Gamma globulin
D. Transferrin
4. What is the normal concentration of total plasma protein?
A. 2–4 g/dl
B. 6–8 g/dl
C. 10–12 g/dl
D. 15–20 g/dl
5. Which plasma protein acts as an antibody?
A. Alpha globulin
B. Beta globulin
C. Gamma globulin
D. Fibrinogen
6. What is the definition of an enzyme?
A. A lipid transporter
B. A protein catalyst
C. A hormone regulator
D. A carbohydrate polymer
7. Which enzyme is most specific for liver disease?
A. AST
B. ALT
C. CK
D. LDH
8. Which enzyme rises first after myocardial infarction?
A. CK-MB
B. LDH
C. ALT
D. ALP

9. What is the normal AST/ALT ratio range?
- A. 0.1–0.3
 - B. 0.7–1.4
 - C. 2–3
 - D. 5–10
10. Which enzyme is a marker of chronic alcoholism?
- A. ALP
 - B. GGT
 - C. CK
 - D. LDH
11. Which test is commonly used for newborn metabolic screening?
- A. Urine dipstick
 - B. Heel-prick blood spot
 - C. Chest X-ray
 - D. Ultrasound
12. Which disorder is caused by deficiency of phenylalanine hydroxylase?
- A. Galactosemia
 - B. PKU
 - C. G6PD deficiency
 - D. Von Gierke disease
13. Which food can trigger hemolysis in G6PD deficiency?
- A. Rice
 - B. Fava beans
 - C. Milk
 - D. Wheat
14. Which metabolic disorder is due to defective branched-chain ketoacid dehydrogenase?
- A. PKU
 - B. Maple Syrup Urine Disease
 - C. Galactosemia
 - D. Fabry disease
15. Which method directly detects mutations in metabolic genes?
- A. Enzyme assay
 - B. Genetic testing
 - C. Mass spectrometry
 - D. Ultracentrifugation
16. Which cardiac marker is the most specific for myocardial necrosis?
- A. CK-MB
 - B. Troponin I
 - C. LDH
 - D. Myoglobin
17. Which marker rises earliest after myocardial infarction?
- A. Myoglobin
 - B. Troponin I
 - C. CK-MB
 - D. LDH

18. Which cardiac marker remains elevated the longest after MI?
- A. Myoglobin
 - B. CK-MB
 - C. Troponin
 - D. LDH
19. Which peptide is used to evaluate congestive heart failure?
- A. BNP
 - B. CK
 - C. LDH
 - D. AST
20. Which cardiac marker peaks at 24 hours after MI?
- A. CK-MB
 - B. Troponin I
 - C. LDH
 - D. Myoglobin
21. Why does albumin deficiency often lead to edema?
- A. It increases blood pressure
 - B. It lowers plasma oncotic pressure
 - C. It raises calcium levels
 - D. It enhances clotting
22. Why are gamma globulins important in clinical practice?
- A. They regulate osmotic pressure
 - B. They act as antibodies
 - C. They transport lipids
 - D. They bind copper
23. Why is fibrinogen considered an acute phase protein?
- A. It decreases during inflammation
 - B. It increases to promote clot formation
 - C. It transports hormones
 - D. It buffers plasma pH
24. Why might serum protein electrophoresis be ordered for a patient?
- A. To measure blood glucose
 - B. To detect abnormal protein bands
 - C. To assess kidney filtration rate
 - D. To check electrolyte balance
25. Why does hypergammaglobulinemia occur in chronic infections?
- A. Increased antibody production
 - B. Reduced albumin synthesis
 - C. Loss of fibrinogen
 - D. Dehydration
26. Why are ALT and AST useful in liver disease diagnosis?
- A. They measure bilirubin directly
 - B. They leak into plasma when hepatocytes are damaged
 - C. They regulate clotting factors
 - D. They transport iron

27. Why is CK-MB more specific for myocardial infarction than total CK?

- A. CK-MB is only found in skeletal muscle
- B. CK-MB is concentrated in heart muscle
- C. CK-MB is a liver enzyme
- D. CK-MB is unaffected by trauma

28. Why is GGT a marker of alcohol abuse?

- A. Alcohol induces enzyme proliferation in the liver
- B. Alcohol decreases enzyme clearance
- C. Alcohol binds directly to GGT
- D. Alcohol reduces bilirubin levels

29. Why is LDH not specific for one disease?

- A. LDH is present in many tissues
- B. LDH is only found in the pancreas
- C. LDH is destroyed in plasma
- D. LDH is a hormone

30. Why is enzyme activity measured instead of enzyme concentration?

- A. Enzymes are unstable in plasma
- B. Activity reflects functional presence
- C. Concentration is always constant
- D. Activity is unrelated to disease

31. Why is newborn screening critical for metabolic disorders?

- A. It prevents genetic inheritance
- B. It detects disease before symptoms appear
- C. It cures metabolic disorders
- D. It replaces genetic counseling

32. Why does G6PD deficiency cause hemolysis after oxidative stress?

- A. RBCs lack glutathione protection
- B. RBCs produce excess bilirubin
- C. RBCs increase enzyme activity
- D. RBCs absorb too much oxygen

33. Why is PKU dangerous if untreated?

- A. Phenylalanine accumulates and damages the brain
- B. Tyrosine levels increase
- C. Insulin resistance develops
- D. Hemoglobin is destroyed

34. Why is MSUD named “Maple Syrup Urine Disease”?

- A. Urine smells sweet due to branched-chain ketoacid buildup
- B. Patients crave maple syrup
- C. It affects carbohydrate metabolism
- D. It is treated with maple syrup

35. Why is genetic testing useful in metabolic disorders?

- A. It identifies enzyme activity directly
- B. It confirms mutations causing disease
- C. It measures blood glucose
- D. It prevents inheritance

36. Why is troponin I considered the gold standard for MI diagnosis?

- A. It is only found in skeletal muscle
- B. It is highly specific to cardiac tissue
- C. It peaks earliest after infarction
- D. It is cleared rapidly

37. Why is myoglobin not specific for MI?

- A. It is also abundant in skeletal muscle
- B. It remains elevated for weeks
- C. It is only found in kidneys
- D. It binds calcium

38. Why is CK-MB useful in detecting reinfarction?

- A. It rises and falls quickly
- B. It stays elevated for 14 days
- C. It is unaffected by trauma
- D. It is only found in brain tissue

39. Why is BNP measured in heart failure patients?

- A. BNP reflects ventricular stress and fluid overload
- B. BNP measures oxygen saturation
- C. BNP is a clotting factor
- D. BNP is a liver enzyme

40. Why do clinicians use multiple cardiac markers together?

- A. To confirm diagnosis with different time courses
- B. To measure cholesterol
- C. To detect anemia
- D. To monitor kidney function

41. A patient with nephrotic syndrome shows low serum albumin and edema. Which mechanism explains this?

- A. Increased plasma oncotic pressure
- B. Decreased plasma oncotic pressure
- C. Increased fibrinogen synthesis
- D. Increased gamma globulin levels

42. Electrophoresis shows a monoclonal spike in the gamma region. Which condition is most likely?

- A. Multiple myeloma
- B. Chronic infection
- C. Nephrotic syndrome
- D. Cirrhosis

43. A patient with hemolytic anemia has low haptoglobin. Why?

- A. Haptoglobin is consumed binding free hemoglobin
- B. Haptoglobin synthesis is reduced in the liver
- C. Haptoglobin is excreted in urine
- D. Haptoglobin is converted to fibrinogen

44. Which abnormality would most likely cause bisalbuminemia?

- A. Genetic mutation in albumin gene
- B. Increased fibrinogen synthesis
- C. Loss of transferrin
- D. Elevated ceruloplasmin

45. A patient with cirrhosis shows β - γ bridging on electrophoresis. Which protein is responsible?
A. IgA
B. IgG
C. Albumin
D. Fibrinogen
46. A patient has ALT > AST (ratio < 0.8). Which condition is most likely?
A. Alcoholic hepatitis
B. Acute hepatocellular injury
C. Cirrhosis
D. NASH
47. CK-MB rises and falls quickly after MI. Why is this clinically useful?
A. It helps detect reinfarction
B. It confirms chronic liver disease
C. It rules out pancreatitis
D. It measures bilirubin clearance
48. A patient with bone pain shows elevated ALP but normal GGT. What is the likely source?
A. Bone disease
B. Liver disease
C. Pancreatitis
D. Muscle injury
49. A statin-treated patient develops muscle pain and CK > 14,000 U/L. What is the diagnosis?
A. Rhabdomyolysis
B. Viral hepatitis
C. Cirrhosis
D. Pancreatitis
50. LDH isoenzyme LDH1 is elevated. Which tissue is most likely damaged?
A. Heart
B. Liver
C. Skeletal muscle
D. Bone
51. A newborn with jaundice and cataracts is diagnosed with galactosemia. Which enzyme is deficient?
A. Galactose-1-phosphate uridylyltransferase (GALT)
B. Phenylalanine hydroxylase
C. Branched-chain ketoacid dehydrogenase
D. G6PD
52. A child with hypoglycemia during fasting is suspected of Von Gierke disease. Which defect explains this?
A. Inability to release glucose from glycogen
B. Excess insulin secretion
C. Defective fatty acid oxidation
D. Lack of lactase enzyme
53. A patient with G6PD deficiency develops hemolysis after eating fava beans. What is the biochemical basis?
A. Reduced NADPH $\rightarrow$ impaired glutathione defense
B. Increased bilirubin synthesis
C. Excess tyrosine production
D. Blocked glycogen breakdown

54. A patient with MSUD shows neurotoxicity. Which metabolites accumulate?

- A. Branched-chain amino acids
- B. Aromatic amino acids
- C. Simple sugars
- D. Fatty acids

55. A family history of PKU increases risk in offspring. Which preventive measure is most appropriate?

- A. Genetic counseling
- B. Liver biopsy
- C. Enzyme replacement therapy
- D. Bone marrow transplant

56. A patient presents with chest pain. Troponin is normal at 2 hours but elevated at 12 hours. What does this indicate?

- A. Troponin rises later after MI
- B. Troponin is not cardiac-specific
- C. CK-MB is more reliable
- D. BNP should be measured instead

57. A patient with renal failure shows elevated troponin T but normal troponin I. Why?

- A. Troponin T is less specific, also found in muscle
- B. Troponin I is excreted in urine
- C. Troponin I is unstable in plasma
- D. Troponin T is unaffected by renal clearance

58. BNP is elevated in heart failure. What does this reflect?

- A. Ventricular wall stretch and fluid overload
- B. Increased liver enzyme activity
- C. Elevated skeletal muscle breakdown
- D. Increased bilirubin clearance

59. Myoglobin rises early but is nonspecific. Why is it still clinically useful?

- A. High negative predictive value for MI
- B. It remains elevated for weeks
- C. It is specific to cardiac tissue
- D. It measures cholesterol

60. A patient with suspected reinfarction shows a second rise in CK-MB. What does this confirm?

- A. A new myocardial infarction
- B. Chronic liver disease
- C. Pancreatitis
- D. Muscle dystrophy

61. A patient with severe burns shows massive plasma loss. Which protein deficiency explains the edema?

- A. Albumin
- B. Fibrinogen
- C. Gamma globulin
- D. Transferrin

62. A patient with Wilson disease has low ceruloplasmin. What is the clinical consequence?

- A. Copper accumulation in tissues
- B. Iron deficiency anemia
- C. Increased fibrinogen
- D. Loss of antibodies

63. A patient with hemolytic anemia shows decreased haptoglobin. What is the mechanism?
- A. Binding of free hemoglobin in plasma
 - B. Reduced liver synthesis
 - C. Increased fibrinogen degradation
 - D. Excess albumin excretion
64. A patient with nephrotic syndrome shows increased α_2 -macroglobulin. Why?
- A. It compensates for protein loss in urine
 - B. It transports copper
 - C. It binds free hemoglobin
 - D. It acts as antibody
65. A patient with chronic inflammation shows elevated C-reactive protein. What does this indicate?
- A. Acute phase response
 - B. Albumin deficiency
 - C. Fibrinogen loss
 - D. Gamma globulin deficiency
66. A patient with chest pain has CK-MB elevated at 6 hours. What does this suggest?
- A. Myocardial infarction
 - B. Viral hepatitis
 - C. Bone disease
 - D. Pancreatitis
67. A patient with obstructive jaundice shows elevated ALP and GGT. What is the source?
- A. Hepatobiliary tract
 - B. Bone
 - C. Muscle
 - D. Pancreas
68. A patient with acute pancreatitis shows elevated lipase. Why is lipase more useful than amylase?
- A. Longer half-life and higher specificity
 - B. Lower plasma concentration
 - C. Faster clearance
 - D. Produced in salivary glands
69. A patient with Paget's disease shows elevated ALP but normal bilirubin. What does this indicate?
- A. Bone origin of ALP
 - B. Liver origin of ALP
 - C. Pancreatic origin of ALP
 - D. Muscle origin of ALP
70. A patient with suspected rhabdomyolysis shows CK > 10,000 U/L. Which complication must be monitored?
- A. Renal failure due to myoglobin
 - B. Cirrhosis
 - C. Pancreatitis
 - D. Osteomalacia
71. A newborn heel-prick test shows low G6PD activity. What advice should be given to parents?
- A. Avoid oxidative drugs and fava beans
 - B. Increase protein intake
 - C. Avoid lactose-containing foods
 - D. Give iron supplements

72. A child with PKU is placed on a special diet. What nutrient must be restricted?

- A. Phenylalanine
- B. Tyrosine
- C. Galactose
- D. Glucose

73. A patient with galactosemia must avoid which food?

- A. Milk and dairy products
- B. Wheat
- C. Rice
- D. Meat

74. A patient with Von Gierke disease presents with hypoglycemia. Which metabolic pathway is defective?

- A. Glycogenolysis (release of glucose)
- B. Lipid oxidation
- C. Protein synthesis
- D. Urea cycle

75. A patient with Fabry disease shows kidney failure. Which enzyme is deficient?

- A. α -galactosidase A
- B. Phenylalanine hydroxylase
- C. G6PD
- D. Galactose-1-phosphate uridylyltransferase

76. A patient with chest pain has elevated troponin but normal CK-MB. What does this suggest?

- A. Microinfarction (small necrosis)
- B. Skeletal muscle injury
- C. Pancreatitis
- D. Bone disease

77. A patient with heart failure shows elevated NT-proBNP. What does this reflect?

- A. Ventricular wall stress
- B. Liver damage
- C. Skeletal muscle breakdown
- D. Kidney filtration rate

78. A patient with suspected MI shows elevated LDH1 after 24 hours. What does this confirm?

- A. Myocardial infarction
- B. Bone disease
- C. Pancreatitis
- D. Cirrhosis

79. A patient with renal failure shows elevated troponin T but normal troponin I. What is the interpretation?

- A. Troponin T is less specific, also found in muscle
- B. Troponin I is excreted in urine
- C. Troponin I is unstable in plasma
- D. Troponin T is unaffected by clearance

80. A patient with chest pain shows myoglobin elevated at 2 hours but normal troponin. What is the best next step?

- A. Repeat troponin after 6–12 hours
- B. Diagnose MI immediately
- C. Rule out cardiac disease
- D. Order liver enzymes

81. A patient with hypoalbuminemia and edema is being evaluated. Which intervention is most appropriate?
A. Increase dietary protein intake
B. Restrict water intake only
C. Stop all medications
D. Ignore unless severe
82. A patient with suspected multiple myeloma shows a monoclonal spike in gamma globulin. Which test should be prioritized?
A. Serum protein electrophoresis
B. Blood glucose test
C. Liver function test
D. Electrolyte panel
83. A patient with Wilson disease is suspected. Which plasma protein measurement is most useful?
A. Ceruloplasmin
B. Albumin
C. Fibrinogen
D. Haptoglobin
84. A patient with nephrotic syndrome shows proteinuria. Which plasma protein loss is most clinically significant?
A. Albumin
B. Fibrinogen
C. Gamma globulin
D. Transferrin
85. A patient with chronic inflammation shows elevated CRP. Which clinical decision is most appropriate?
A. Monitor as a marker of inflammation
B. Ignore the result
C. Stop all protein intake
D. Diagnose liver disease immediately
86. A patient with chest pain has elevated CK-MB and troponin. Which marker should be considered most specific for MI?
A. Troponin I
B. CK-MB
C. LDH
D. AST
87. A patient with obstructive jaundice shows elevated ALP and GGT. Which conclusion is most valid?
A. Hepatobiliary obstruction
B. Bone disease
C. Muscle injury
D. Viral infection
88. A patient with suspected rhabdomyolysis shows CK > 10,000 U/L. Which immediate evaluation is most critical?
A. Renal function tests
B. Liver enzymes
C. Bone density scan
D. Blood glucose
89. A patient with Paget's disease shows elevated ALP but normal GGT. Which source is most likely?
A. Bone
B. Liver
C. Pancreas

D. Muscle

90. A patient with suspected acute pancreatitis shows elevated lipase. Which evaluation is most appropriate?

A. Confirm diagnosis with lipase (more specific than amylase)

B. Ignore enzyme results

C. Use CK only

D. Order bone scan

91. A newborn screening shows G6PD deficiency. Which preventive measure is most important?

A. Avoid oxidative drugs and foods

B. Increase carbohydrate intake

C. Restrict protein intake

D. Give iron supplements

92. A child with PKU is diagnosed early. Which evaluation is most critical for management?

A. Dietary restriction of phenylalanine

B. Increase tyrosine intake only

C. Stop all protein intake

D. Use antibiotics

93. A patient with galactosemia is evaluated. Which dietary change is most appropriate?

A. Avoid milk and dairy products

B. Avoid wheat

C. Avoid rice

D. Avoid meat

94. A patient with Von Gierke disease presents with hypoglycemia. Which evaluation is most critical?

A. Assess glycogenolysis defect

B. Check calcium levels

C. Measure bilirubin

D. Evaluate bone density

95. A patient with Fabry disease shows kidney failure. Which enzyme deficiency should be confirmed?

A. α -galactosidase A

B. Phenylalanine hydroxylase

C. G6PD

D. Galactose-1-phosphate uridylyltransferase

96. A patient with chest pain shows elevated troponin but normal CK-MB. Which evaluation is most appropriate?

A. Diagnose microinfarction

B. Rule out MI completely

C. Diagnose pancreatitis

D. Order bone scan

97. A patient with heart failure shows elevated NT-proBNP. Which conclusion is most valid?

A. Ventricular wall stress and fluid overload

B. Liver disease

C. Muscle injury

D. Kidney disease

98. A patient with suspected MI shows elevated LDH1 after 24 hours. Which evaluation is most appropriate?

A. Confirm myocardial infarction

B. Diagnose bone disease

C. Diagnose cirrhosis

D. Rule out cardiac disease

99. A patient with renal failure shows elevated troponin T but normal troponin I. Which evaluation is most valid?
- A. Troponin T is less specific, also found in muscle
 - B. Troponin I is unstable
 - C. Troponin I is excreted in urine
 - D. Troponin T is unaffected by clearance
100. A patient with chest pain shows myoglobin elevated at 2 hours but normal troponin. Which evaluation is most appropriate?
- A. Repeat troponin after 6–12 hours
 - B. Diagnose MI immediately
 - C. Rule out cardiac disease
 - D. Order liver enzymes
101. A researcher wants to design a test to detect hypoalbuminemia. Which method should they choose?
- A. Serum protein electrophoresis
 - B. Albumin concentration assay
 - C. Blood glucose test
 - D. Lipid profile
102. A clinician wants to monitor inflammation. Which plasma protein should they measure?
- A. C-reactive protein
 - B. Albumin
 - C. Transferrin
 - D. Fibrinogen
103. A scientist wants to study copper metabolism. Which plasma protein should they analyze?
- A. Ceruloplasmin
 - B. Albumin
 - C. Haptoglobin
 - D. Fibrinogen
104. A lab wants to confirm multiple myeloma. Which plasma protein band should they focus on?
- A. Gamma globulin
 - B. Albumin
 - C. Fibrinogen
 - D. Alpha-1 globulin
105. A doctor wants to evaluate nephrotic syndrome. Which plasma protein should be measured in urine?
- A. Albumin
 - B. Fibrinogen
 - C. Gamma globulin
 - D. Transferrin
106. A hospital wants to design a protocol for MI diagnosis. Which enzyme should be prioritized?
- A. Troponin I
 - B. CK-MB
 - C. LDH
 - D. ALT
107. A researcher wants to differentiate bone vs liver disease using enzymes. Which combination is best?
- A. ALP and GGT
 - B. CK and LDH
 - C. ALT and AST
 - D. Lipase and Amylase

108. A clinician wants to confirm pancreatitis. Which enzyme should they measure?

- A. Lipase
- B. CK
- C. ALT
- D. LDH

109. A scientist wants to design a test for chronic alcoholism. Which enzyme marker should they use?

- A. GGT
- B. CK-MB
- C. LDH1
- D. ALP

110. A doctor wants to evaluate muscle injury. Which enzyme should be measured?

- A. CK
- B. ALT
- C. ALP
- D. GGT

111. A newborn screening program wants to detect PKU. Which test should they design?

- A. Heel-prick blood spot for phenylalanine
- B. Liver biopsy
- C. Bone marrow test
- D. Urine dipstick

112. A clinician wants to prevent hemolysis in G6PD deficiency. What advice should they create?

- A. Avoid oxidative drugs and foods
- B. Increase protein intake
- C. Restrict carbohydrates
- D. Take iron supplements

113. A genetic counselor wants to design a prevention plan for galactosemia. Which recommendation is best?

- A. Avoid milk and dairy products
- B. Avoid wheat
- C. Avoid rice
- D. Avoid meat

114. A researcher wants to design a therapy for Fabry disease. Which approach is most appropriate?

- A. Enzyme replacement therapy with α -galactosidase A
- B. Increase dietary protein
- C. Restrict glucose intake
- D. Use antibiotics

115. A doctor wants to design a screening for Von Gierke disease. Which parameter should be measured?

- A. Blood glucose during fasting
- B. Serum calcium
- C. Plasma bilirubin
- D. Serum creatinine

116. A hospital wants to design a rapid MI test. Which marker should they use for early detection?

- A. Myoglobin
- B. Troponin I
- C. CK-MB
- D. LDH

117. A clinician wants to confirm MI after 24 hours. Which marker should they rely on?

- A. Troponin I
- B. CK-MB
- C. LDH1
- D. BNP

118. A researcher wants to design a test for heart failure. Which marker should they measure?

- A. NT-proBNP
- B. CK
- C. ALT
- D. LDH

119. A doctor wants to design a protocol for reinfarction detection. Which marker is most useful?

- A. CK-MB
- B. Troponin T
- C. LDH
- D. BNP

120. A lab wants to design a panel combining early and late MI markers. Which combination is best?

- A. Myoglobin + Troponin I
- B. ALT + AST
- C. CK + LDH
- D. ALP + GGT

121. Why is albumin considered unique among plasma proteins?

- A. It is a glycoprotein
- B. It is not a glycoprotein
- C. It is synthesized in plasma cells
- D. It is an acute phase protein

122. Why does hypoalbuminemia reduce calcium levels in plasma?

- A. Albumin binds calcium normally
- B. Albumin increases calcium synthesis
- C. Albumin prevents calcium excretion
- D. Albumin converts calcium to fibrin

123. Why is transferrin important in plasma?

- A. It transports iron to tissues
- B. It transports copper
- C. It acts as antibody
- D. It regulates osmotic pressure

124. Why does fibrinogen appear as an extra band in plasma electrophoresis?

- A. Plasma contains fibrinogen unlike serum
- B. Serum contains more fibrinogen
- C. Fibrinogen migrates with albumin
- D. Fibrinogen is absent in plasma

125. Why are acute phase proteins clinically significant?

- A. They change rapidly in inflammation
- B. They regulate osmotic pressure
- C. They transport hormones
- D. They act as antibodies

126. Why is ALT more specific for liver disease than AST?
- A. ALT is concentrated mainly in hepatocytes
 - B. ALT is found in many tissues
 - C. ALT is unstable in plasma
 - D. ALT is a muscle enzyme
127. Why is CK-MB used in MI diagnosis?
- A. It is more cardiac-specific than total CK
 - B. It is only found in skeletal muscle
 - C. It is a liver enzyme
 - D. It is unaffected by trauma
128. Why is LDH1 elevation significant in MI?
- A. LDH1 is abundant in heart tissue
 - B. LDH1 is only found in bone
 - C. LDH1 is unstable in plasma
 - D. LDH1 is a liver enzyme
129. Why is enzyme activity expressed in IU/L?
- A. It measures substrate conversion per minute
 - B. It measures enzyme concentration directly
 - C. It measures protein synthesis
 - D. It measures bilirubin clearance
130. Why is lipase preferred over amylase in late pancreatitis diagnosis?
- A. Lipase has a longer half-life
 - B. Lipase is less specific
 - C. Lipase is unstable in plasma
 - D. Lipase is produced in salivary glands
131. Why is galactosemia dangerous in infants?
- A. Galactose accumulates and damages the liver
 - B. Galactose increases insulin resistance
 - C. Galactose reduces hemoglobin
 - D. Galactose prevents calcium absorption
132. Why is G6PD deficiency more common in males?
- A. It is X-linked
 - B. It is autosomal dominant
 - C. It is autosomal recessive
 - D. It is mitochondrial
133. Why is MSUD rapidly fatal if untreated?
- A. Neurotoxicity from branched-chain amino acids
 - B. Hypoglycemia from glycogen breakdown
 - C. Copper accumulation in tissues
 - D. Lack of insulin secretion
134. Why is mass spectrometry used in newborn screening?
- A. It profiles multiple metabolites quickly
 - B. It measures bilirubin directly
 - C. It detects calcium levels
 - D. It measures hemoglobin concentration

135. Why is genetic counseling important in metabolic disorders?

- A. It helps families understand inheritance risks
- B. It cures metabolic disease
- C. It replaces enzyme therapy
- D. It prevents infections

136. Why is troponin both an early and late marker of MI?

- A. Cytosolic troponin is released early, sarcomeric later
- B. Troponin is unstable in plasma
- C. Troponin is only found in skeletal muscle
- D. Troponin is cleared rapidly

137. Why is CK-MB less specific than troponin?

- A. CK-MB is also present in skeletal muscle
- B. CK-MB is unstable in plasma
- C. CK-MB is only found in liver
- D. CK-MB is cleared slowly

138. Why is BNP more stable in plasma than BNP itself?

- A. NT-proBNP is biologically inactive and cleared slower
- B. BNP is more concentrated in plasma
- C. BNP is only found in skeletal muscle
- D. NT-proBNP is unstable

139. Why is myoglobin considered useful despite poor specificity?

- A. It rises earliest after MI
- B. It remains elevated for weeks
- C. It is only found in cardiac tissue
- D. It measures cholesterol

140. Why do clinicians measure multiple cardiac markers together?

- A. Different markers rise and fall at different times
- B. They measure cholesterol
- C. They detect anemia
- D. They monitor kidney function

141. What is the half-life of albumin in plasma?

- A. 2 days
- B. 20 days
- C. 5 days
- D. 12 hours

142. Which plasma protein transports thyroid hormones and retinol?

- A. Albumin
- B. Prealbumin
- C. Transferrin
- D. Ceruloplasmin

143. Which plasma protein binds free hemoglobin to conserve iron?

- A. Haptoglobin
- B. Fibrinogen
- C. Albumin
- D. Gamma globulin

144. Which plasma protein transports copper in the blood?
A. Ceruloplasmin
B. Transferrin
C. Albumin
D. Fibrinogen
145. Which plasma protein increases in nephrotic syndrome?
A. α 2-macroglobulin
B. Albumin
C. Prealbumin
D. Gamma globulin
146. What is the normal reference value for ALT in males?
A. < 20 U/L
B. < 45 U/L
C. < 100 U/L
D. < 200 U/L
147. Which enzyme hydrolyses organic phosphates at high pH?
A. Acid phosphatase
B. Alkaline phosphatase
C. Lipase
D. Amylase
148. Which enzyme is a marker of chronic alcoholism?
A. GGT
B. CK
C. LDH
D. ALT
149. Which enzyme breaks down starch and glycogen to maltose?
A. Lipase
B. Amylase
C. CK
D. LDH
150. Which enzyme catalyzes the reversible conversion of lactate to pyruvate?
A. LDH
B. CK
C. ALT
D. AST
151. Which enzyme deficiency causes PKU?
A. Phenylalanine hydroxylase
B. Galactose-1-phosphate uridylyltransferase
C. α -galactosidase A
D. G6PD
152. Which metabolic disorder is due to defective branched-chain ketoacid dehydrogenase?
A. MSUD
B. PKU
C. Galactosemia
D. Von Gierke disease

153. Which enzyme deficiency causes hemolytic anemia after oxidative stress?

- A. G6PD
- B. ALT
- C. LDH
- D. CK

154. Which enzyme deficiency causes Fabry disease?

- A. α -galactosidase A
- B. Phenylalanine hydroxylase
- C. G6PD
- D. Galactose-1-phosphate uridylyltransferase

155. Which enzyme deficiency causes Von Gierke disease?

- A. Glucose-6-phosphatase
- B. ALT
- C. LDH
- D. CK

156. Which cardiac marker peaks at 12–24 hours after MI?

- A. Troponin I/T
- B. CK-MB
- C. LDH
- D. Myoglobin

157. Which cardiac marker peaks at 24 hours and returns to baseline in 48–72 hours?

- A. CK-MB
- B. Troponin I
- C. LDH
- D. BNP

158. Which cardiac marker rises earliest after MI (1–4 hours)?

- A. Myoglobin
- B. Troponin I
- C. CK-MB
- D. LDH

159. Which peptide is used to evaluate congestive heart failure?

- A. BNP
- B. CK
- C. LDH
- D. AST

160. Which cardiac marker remains elevated for up to 14 days after MI?

- A. Troponin
- B. CK-MB
- C. LDH
- D. Myoglobin

Case 1: A 42-year-old man presents with generalized edema, ascites, and fatigue. He has a long history of chronic hepatitis B infection. On examination, he has pitting edema in both legs, a distended abdomen with shifting dullness, and mild jaundice.

Findings: Serum albumin 1.5 g/dL (low), total protein 5.2 g/dL, urine protein negative, liver ultrasound shows cirrhotic changes, serum protein electrophoresis shows reduced albumin band.

Q1: Which plasma protein deficiency explains the edema? **hypoalbuminemia**

Q2: What is the mechanism behind ascites in this patient?

Case 2: A 35-year-old woman presents with recurrent respiratory infections, fatigue, and poor wound healing. She has no history of renal or liver disease.

Findings: Serum electrophoresis shows markedly reduced gamma globulin fraction, albumin normal, fibrinogen normal. Immunoglobulin quantification reveals low IgG and IgA.

Q1: Which plasma protein deficiency explains the recurrent infections?

Q2: What is the role of this protein in immunity?

Case 3: A 55-year-old man with chronic alcoholism presents with fatigue, hepatomegaly, and spider angiomas.

Findings: Serum protein electrophoresis shows β - γ bridging, IgA elevated, albumin reduced. Liver biopsy confirms cirrhosis.

Q1: Which condition is associated with β - γ bridging?

Q2: Which immunoglobulin is mainly responsible for this pattern?

Case 4: A 58-year-old man presents with acute chest pain radiating to the left arm.

Findings: CK-MB elevated at 6 hours (220 U/L), Troponin I elevated at 12 hours, ECG shows ST elevation in anterior leads.

Q1: What is the diagnosis?

Q2: Why is CK-MB useful for detecting reinfarction?

Case 5: A 10-year-old boy presents with bone pain, bowing of legs, and delayed growth.

Findings: ALP 700 U/L (high), GGT normal, vitamin D deficiency confirmed, X-ray shows widened growth plates.

Q1: What is the source of ALP elevation?

Q2: Which condition is most likely?

Case 6: A 45-year-old man with chronic alcohol use presents with fatigue and mild hepatomegaly.

Findings: GGT 250 U/L (high), ALP mildly elevated, ALT normal.

Q1: Which enzyme is a marker of alcohol abuse?

Q2: Why is GGT elevated in this patient?

Case 7: A newborn is irritable, vomiting, and has poor feeding.

Findings: Heel-prick test shows phenylalanine 25 mg/dL (high), Guthrie test positive, tyrosine low, normal liver function.

Q1: Which disorder is suspected?

Q2: What dietary management is required?

Case 8: A 6-month-old boy presents with hypoglycemia during fasting, hepatomegaly, and lactic acidosis.

Findings: Blood glucose 40 mg/dL, lactate elevated, enlarged liver on ultrasound.

Q1: Which disease is suspected?

Q2: Which enzyme defect causes this condition?

Case 9: A 20-year-old man develops jaundice and dark urine after taking antimalarial drugs.

Findings: Hemoglobin 8 g/dL, reticulocytosis, Heinz bodies in RBCs, indirect bilirubin elevated.

Q1: Which disorder is suspected?

Q2: What is the biochemical defect?

Case 10: A 52-year-old man presents with chest pain for 2 hours.

Findings: Myoglobin 600 ng/mL (elevated), Troponin I normal, ECG borderline.

Q1: What does elevated myoglobin indicate?

Q2: What is the best next step in diagnosis?

Case 11: A 65-year-old woman presents with dyspnea and ankle swelling.

Findings: NT-proBNP 3200 pg/mL (high), echocardiogram shows dilated ventricles, EF 35%.

Q1: What does NT-proBNP reflect?

Q2: Why is NT-proBNP more stable than BNP?

Case 12: A 58-year-old man with chest pain shows CK-MB rise at 6 hours, fall at 48 hours, then a second rise at 72 hours.

Findings: CK-MB 220 U/L, Troponin I elevated, ECG shows new ST changes.

Q1: What does the second rise in CK-MB confirm?

Q2: Why is CK-MB useful for detecting reinfarction?

Matching:

Exercise 1: Match Proteins/Enzymes to Their Function

1. Albumin
2. CK-MB
3. G6PD
4. NT-proBNP
5. Ceruloplasmin

Options:

- A. Maintains oncotic pressure
- B. Marker of myocardial infarction
- C. Protects RBCs from oxidative stress
- D. Reflects ventricular wall stress
- E. Transports copper

Exercise 2: Match Disease to Key Laboratory Finding

1. PKU
2. Von Gierke disease

3. Cirrhosis
4. Acute pancreatitis
5. Myocardial infarction

Options:

- A. High phenylalanine
- B. Hypoglycemia with hepatomegaly
- C. β - γ bridging on electrophoresis
- D. Elevated lipase
- E. Elevated troponin

Exercise 3: Match Clinical Feature to Plasma Component

1. Edema in nephrotic syndrome
2. Recurrent infections
3. Jaundice after primaquine
4. Chest pain with ST elevation
5. Dyspnea with heart failure

Options:

- A. Albumin loss
- B. Gamma globulin deficiency
- C. G6PD deficiency
- D. Troponin rise
- E. NT-proBNP elevation

Exercise 4: Match Enzyme/Marker to Timing

1. Myoglobin
2. CK-MB
3. Troponin I
4. LDH1
5. ALT

Options:

- A. Rises earliest (1–4h)
- B. Peaks at 24h, normal by 72h
- C. Rises at 4–9h, stays up to 14 days
- D. Rises at 12–24h, peaks at 3–5 days
- E. Specific for hepatocellular injury

Exercise 5: Match Disorder to Deficient Protein/Enzyme

1. Wilson disease
2. PKU
3. Fabry disease
4. Nephrotic syndrome
5. Rickets

Options:

- A. Ceruloplasmin
- B. Phenylalanine hydroxylase
- C. α -galactosidase A
- D. Albumin
- E. Vitamin D deficiency $\rightarrow$ ALP rise

Exercise 6: Match Case Finding to Diagnosis

1. β - γ bridging on electrophoresis
2. CK-MB second rise after 72h
3. Infant with cataracts and jaundice
4. Elevated GGT in chronic alcohol use
5. NT-proBNP > 3000 pg/mL

Options:

- A. Cirrhosis
- B. Reinfarction
- C. Galactosemia
- D. Alcohol abuse
- E. Heart failure